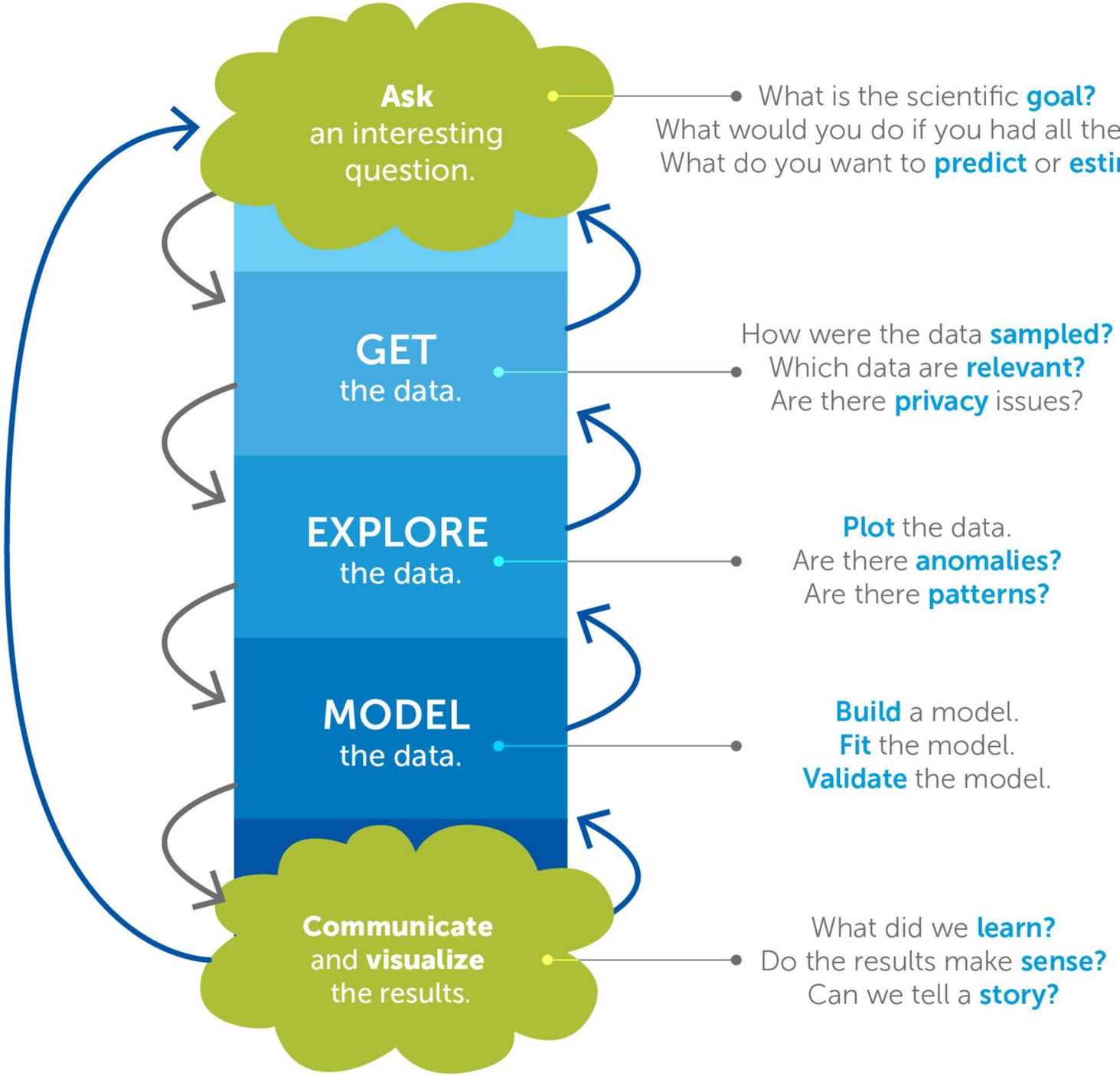
cogs 9 | Week 3

Data visualization

Brendan Tomoschuk



So far:

- What is data science
- Asking good questions
- Ethics & privacy
- Data types and storage
- Getting & cleaning data

Up next:

- Visualizing & exploring data
- Modeling the data (ML)
- Algorithmic bias & fairness
- Data science applications
- Future of data science

Reading(s) 4: *Data Visualization*

Main idea: visualization design choices should be tailored to your audience

Data is Personal: Attitudes and Perceptions of Data Visualization in Rural Pennsylvania

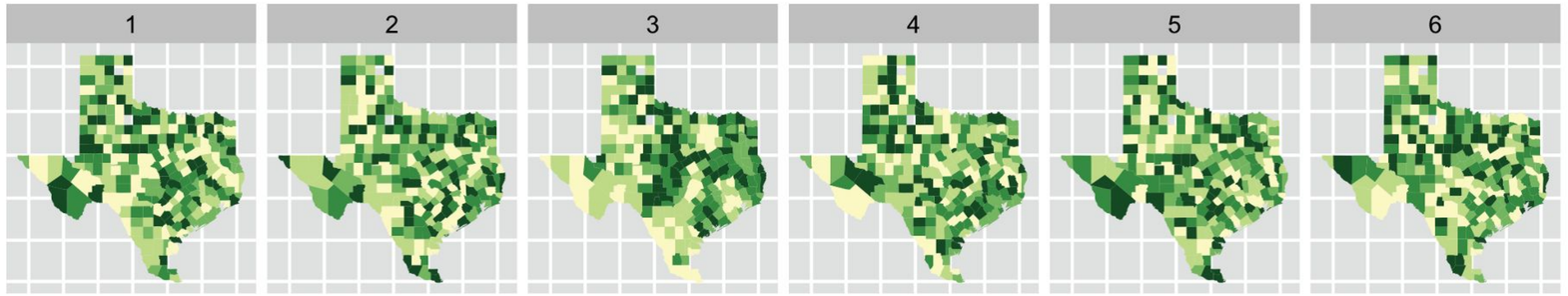
Evan M. Peck
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Graphical Inference for Infovis

Hadley Wickham, Dianne Cook, Heike Hofmann, and Andreas Buja



Main idea: Visualizations are powerful tools for understanding structure in data

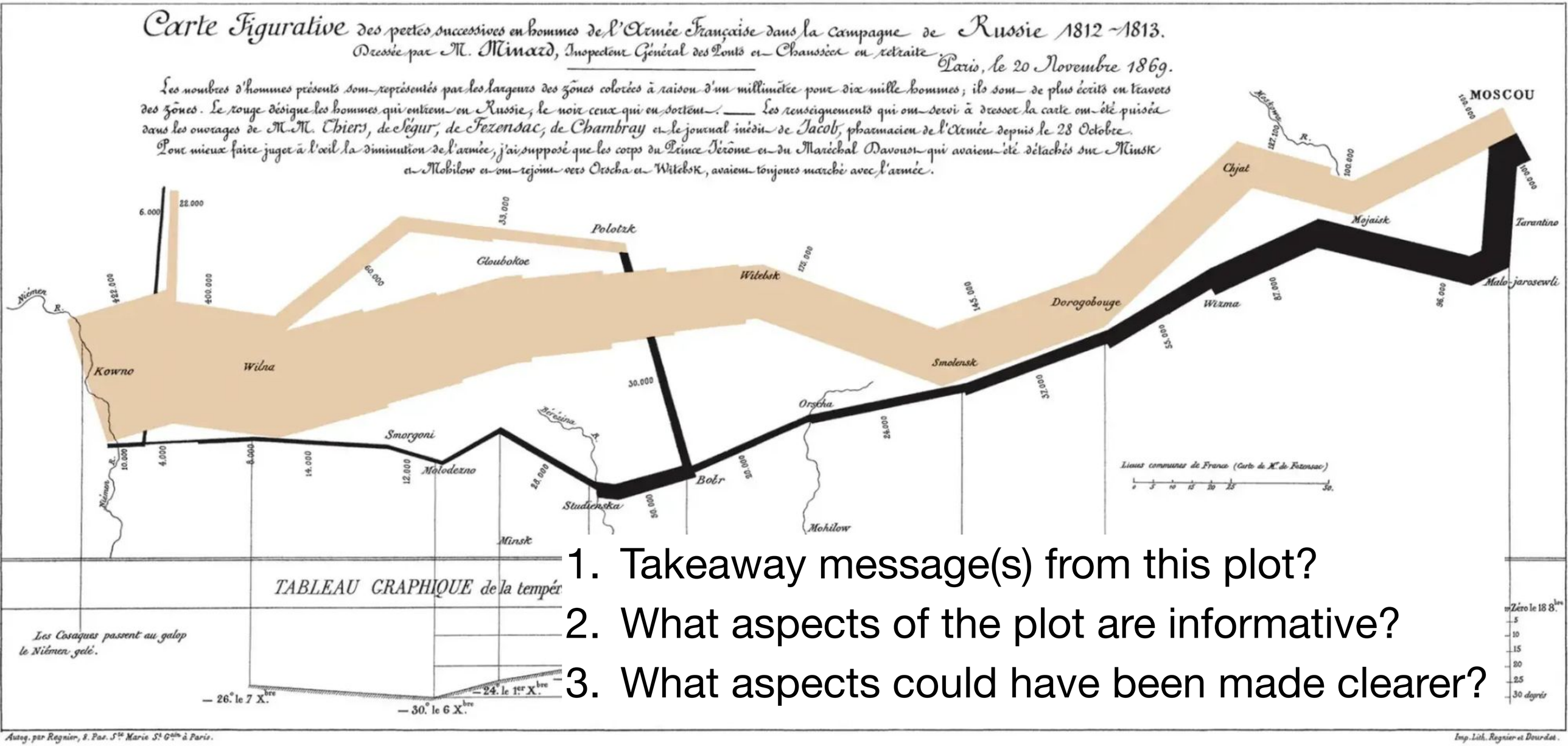
Learning Goals

1. Explain **why data visualizations exist** and when to use them
2. Identify what constitutes an **effective data visualization**
3. Describe **common visual encodings** (position, length, color, size, shape) used in data visualizations
4. Select **appropriate visualization types** based on the data type and question being asked

Why do we visualize data?

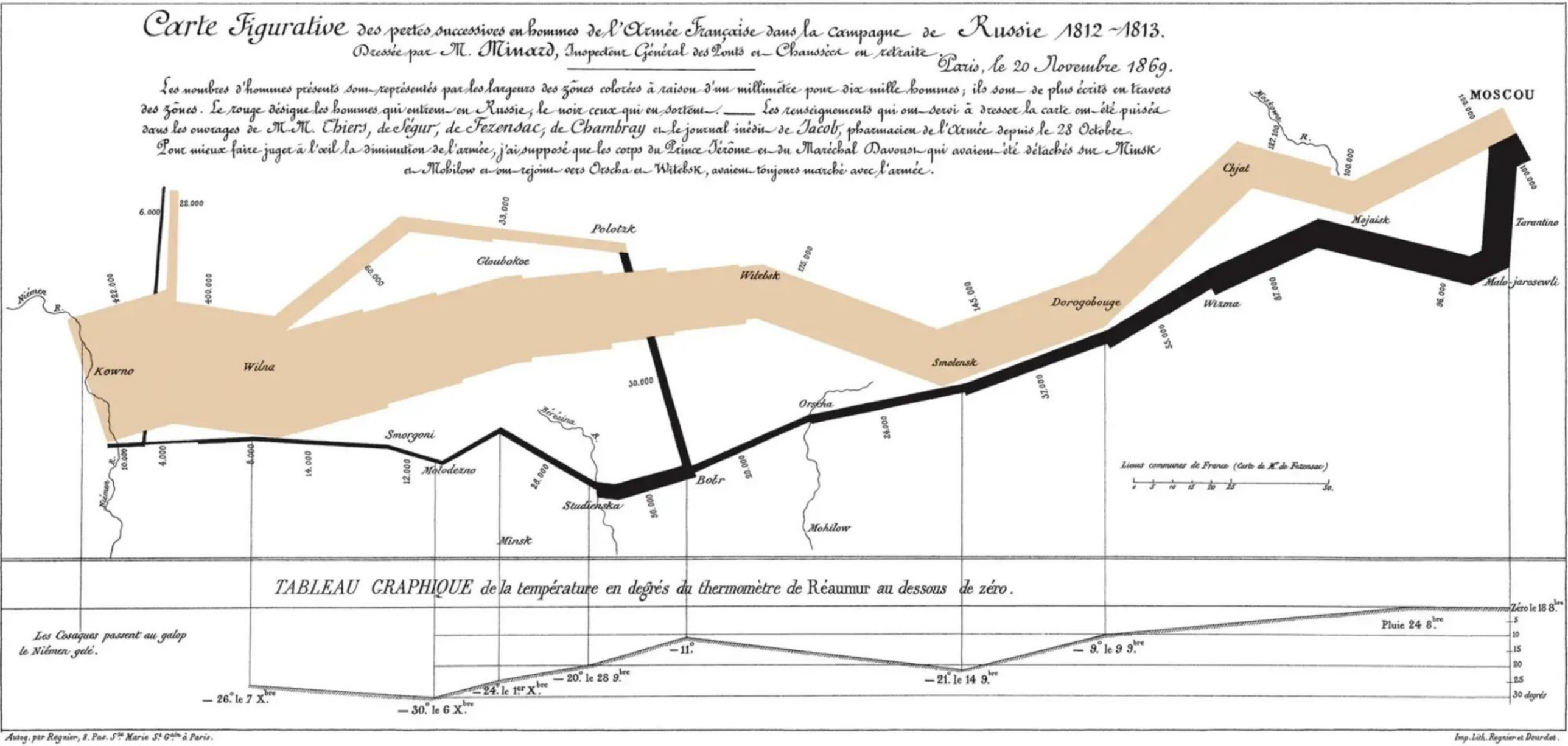
(~2 min with your neighbor)

Napoleon's March to Moscow



1. Takeaway message(s) from this plot?
2. What aspects of the plot are informative?
3. What aspects could have been made clearer?

Visualizations tell a *story*



Reasons to visualize data

1. Raw data is *precise* but often hard to interpret
2. Visualizations reduce cognitive load (mental effort) by making patterns easier to see:
   - Trends over time
   - Group differences
   - Outliers
   - Relationships between variables
3. Some visual features (like position, length, and color) automatically stand out and can help communicate ideas

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course code:

WSTAPX

A visualization is always more helpful than looking at just the raw data

(A) True

(B) False

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course code:

WSTAPX

A visualization is always more helpful than looking at just the raw data

(A) True

(B) False

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course code:
WSTAPX

You are given a table of names and exam scores and want to know what "Amy" got on the exam. A visualization is…

(A) necessary because it will make the data easier to understand

(B) unnecessary because the question can be answered more exactly by looking at the table

(C) necessary because tables are generally difficult for humans to interpret

(D) unnecessary because visualizations are only for large datasets

(E) I'm not sure / I'm confused

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course code:
WSTAPX

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When to NOT use a visualization

Not every question benefits from a visualization!

1. When you or your audience needs ***exact values***
   - Tables are better and often faster for quick lookups!
2. When the **dataset is very small** and easily read/understood

Takeaway: Visualization is a tool to communicate meaning, not a default

A good rule of thumb: Use a visualization only when it makes an insight *faster to see* than the raw data alone

Visual encodings

Visual encodings are **visual properties** used to represent data in a chart

- Quantitative encodings show the *magnitude* of a value
- Qualitative encodings distinguish *categories*

Visual encodings for quantitative data

Quantitative encodings show the *magnitude* of a value

From **most to least** precise:

1. **Position:** value is shown by where a mark is placed along a scale
2. **Length:** value is shown by how long or tall a mark is
3. **Angle:** value is shown by size of an angle
4. **Area:** value is shown by how much 2D space a mark covers
5. **Color:** value is shown by differences in saturation or hue

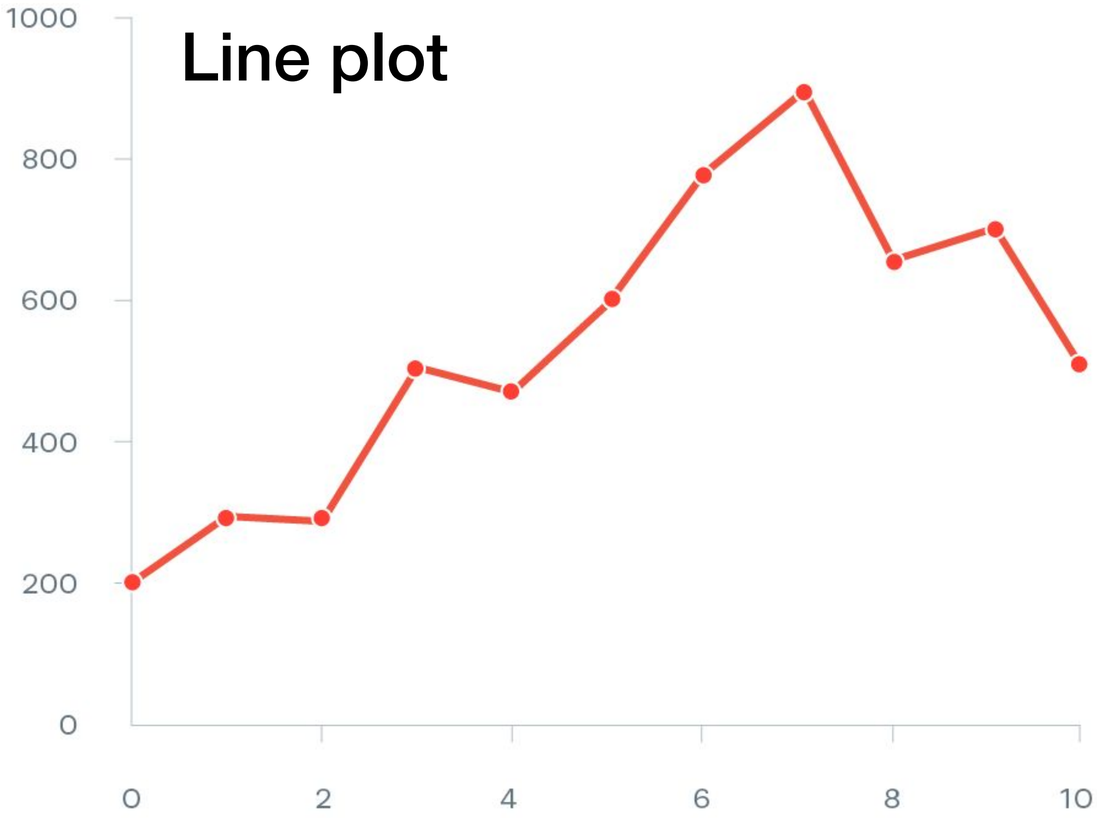
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course code:
WSTAPX

Which quantitative visual encoding is used?



(A) Position

(B) Length

(C) Angle

(D) Area

(E) Color

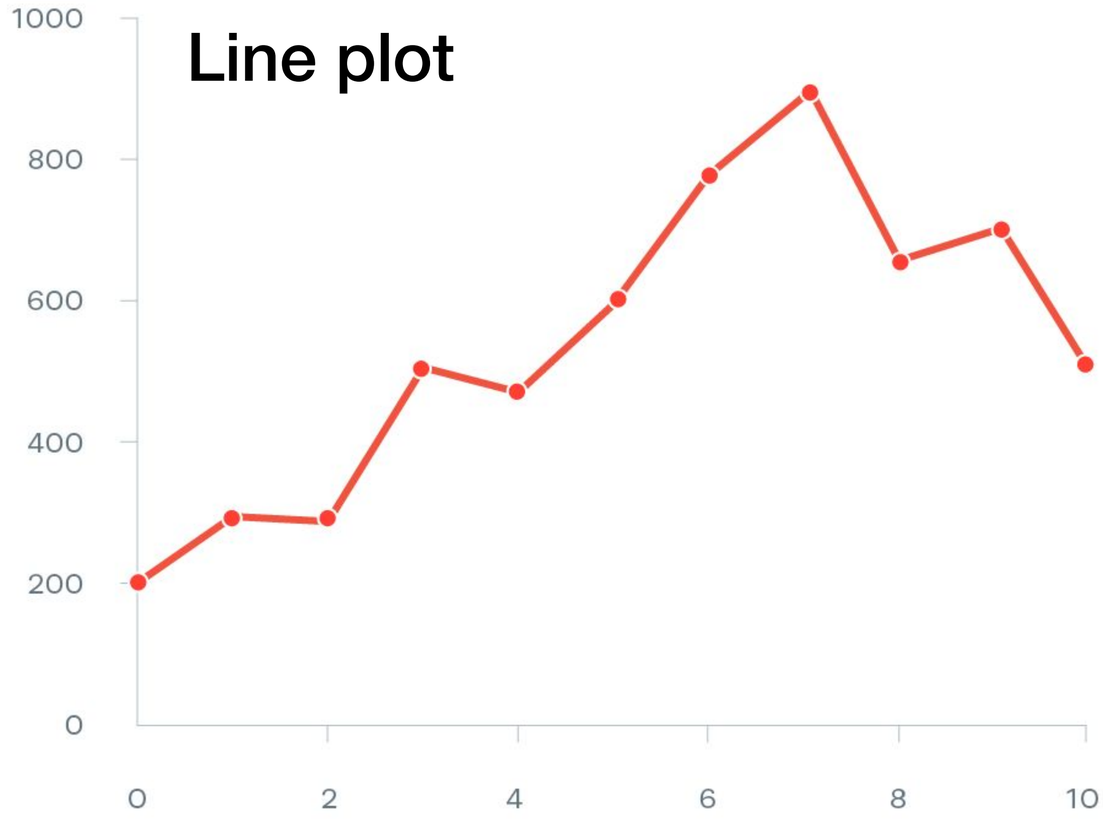
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WSTAPX

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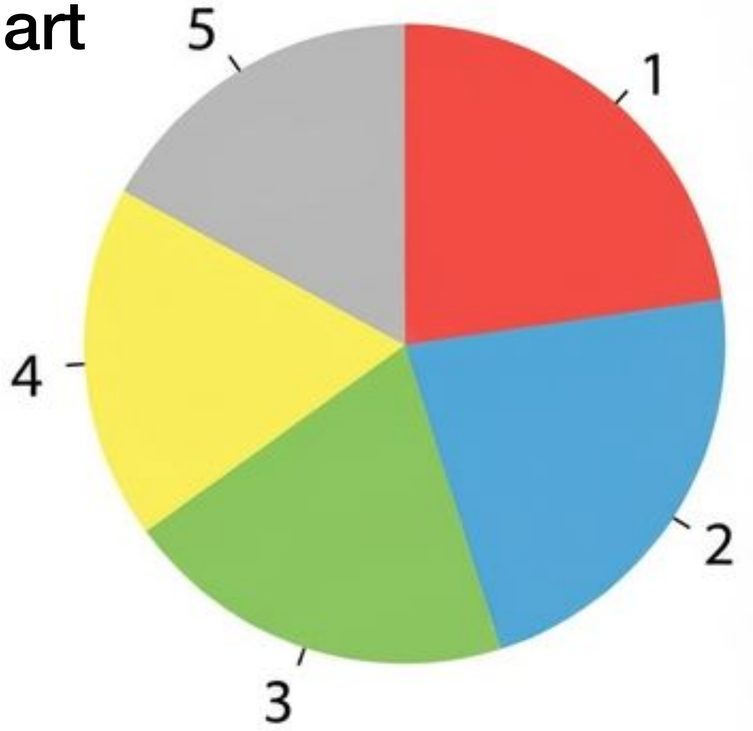
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course code:
WSTAPX

Which quantitative visual encoding(s) are used? *Select all that apply.*

Pie chart



(A) Position

(B) Length

(C) Angle

(D) Area

(E) Color

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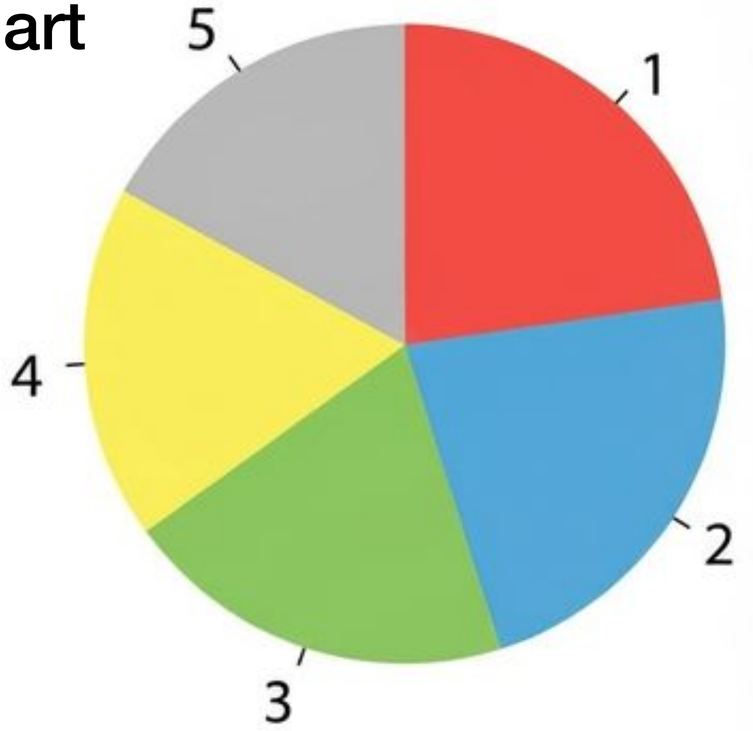
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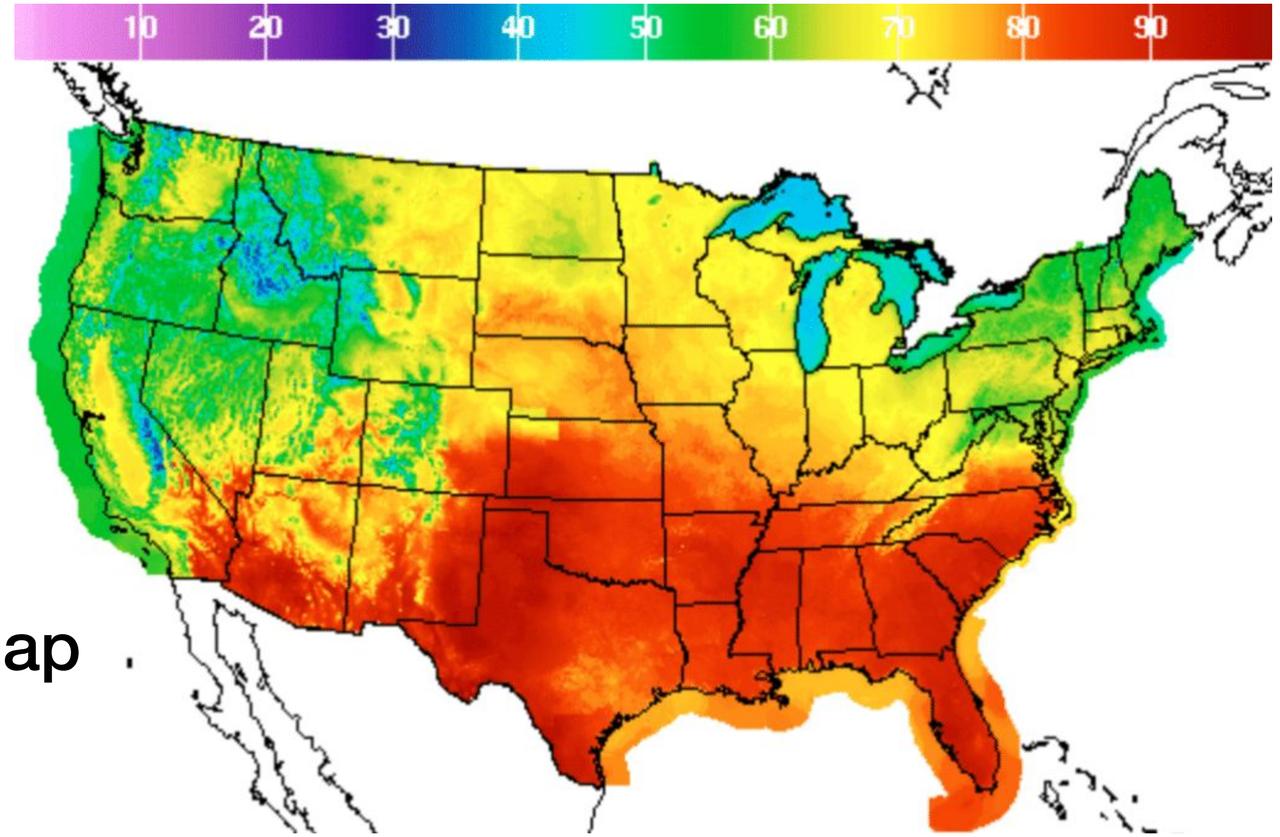
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course code:
WSTAPX

Which quantitative visual encoding is used?



Heatmap

(A) Position

(B) Length

(C) Angle

(D) Area

(E) Color

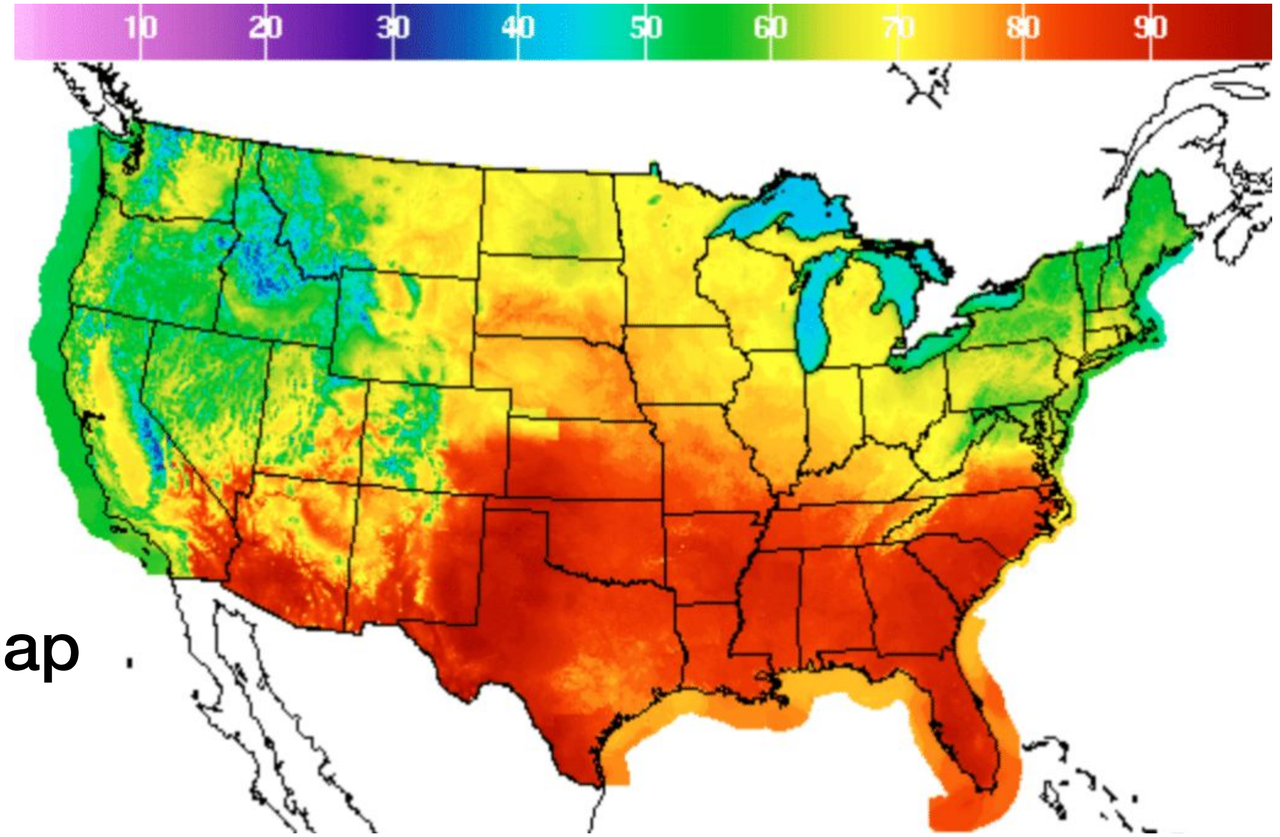
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course code:
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Heatmap

(A) Position

(B) Length

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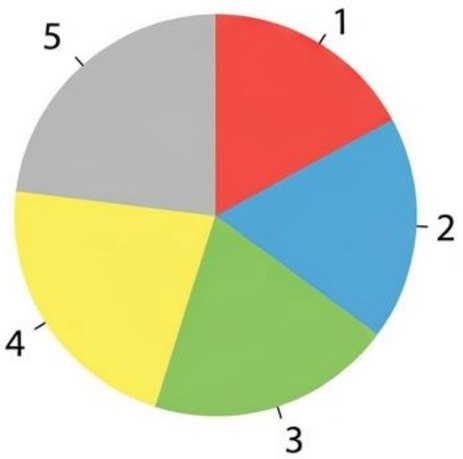
(E) Color

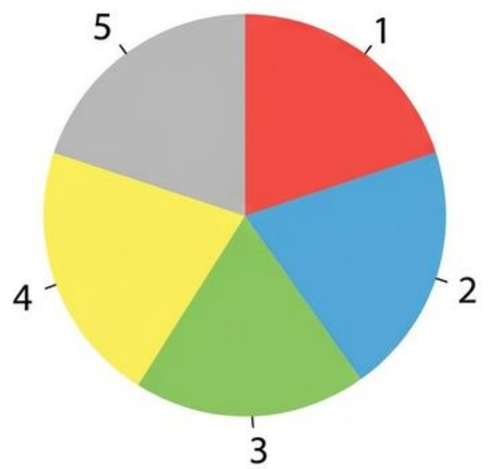
Precision of encodings

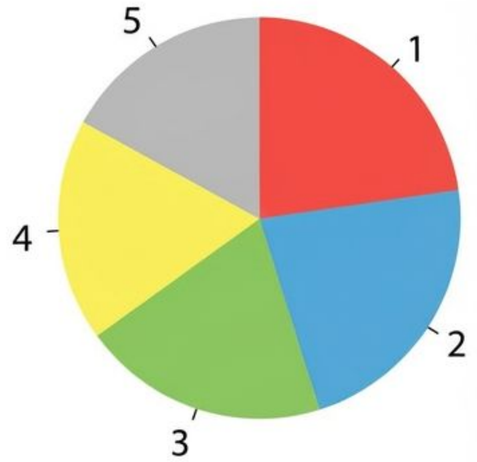
Which category is the largest?

Encoding: Area

Pie charts are *not precise enough* to compare values across categories

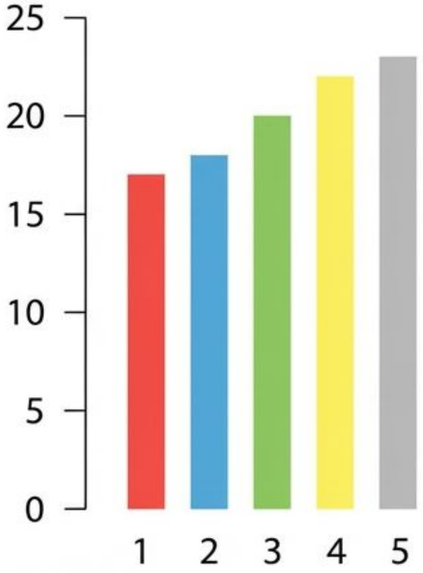


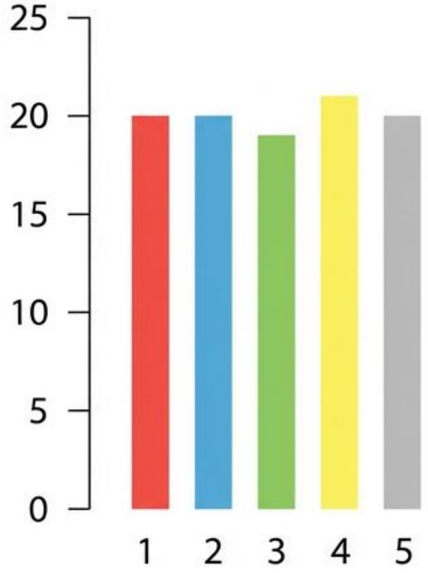


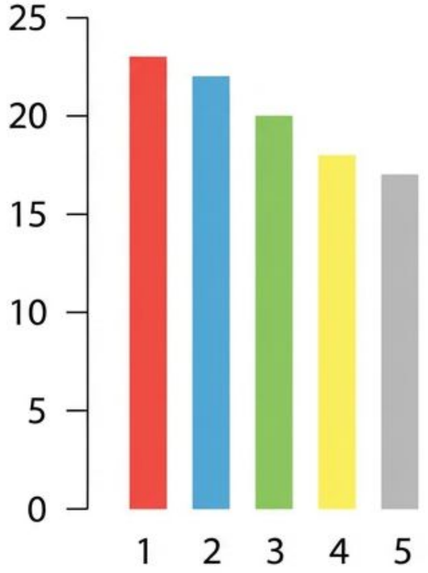


Encoding: Length

Bar charts are a more precise way to compare values across categories







Visual encodings for qualitative data

Qualitative encodings distinguish *categories*

1. **Position:** separate categories by spatial placement or grouping
2. **Color:** different colors for different categories
3. **Shape:** different shapes for different categories
4. **Texture/pattern:** stripes, dots, etc., to distinguish categories

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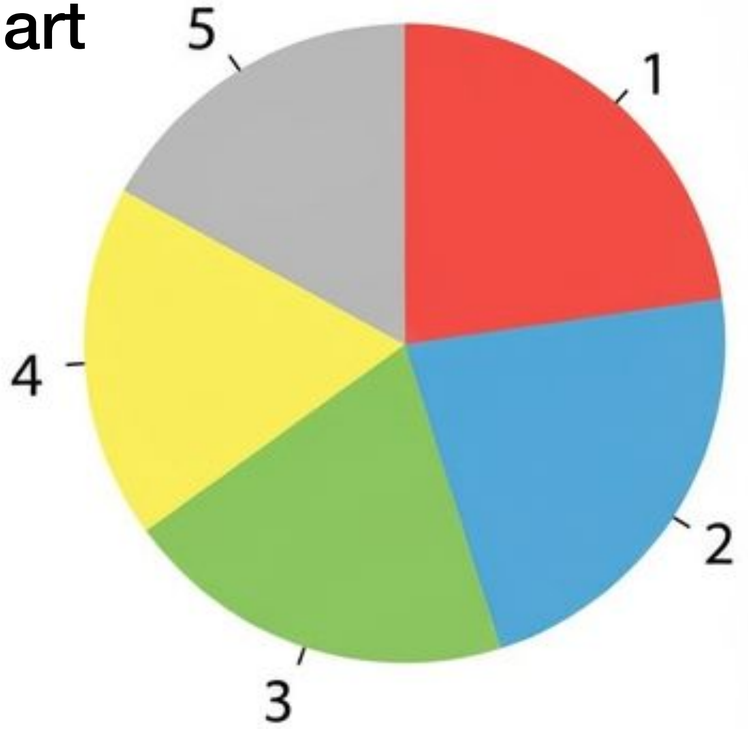
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course code:
WSTAPX

Which qualitative visual encoding is used?

Pie chart



(A) Position

(B) Color

(C) Shape

(D) Angle

(E) Texture

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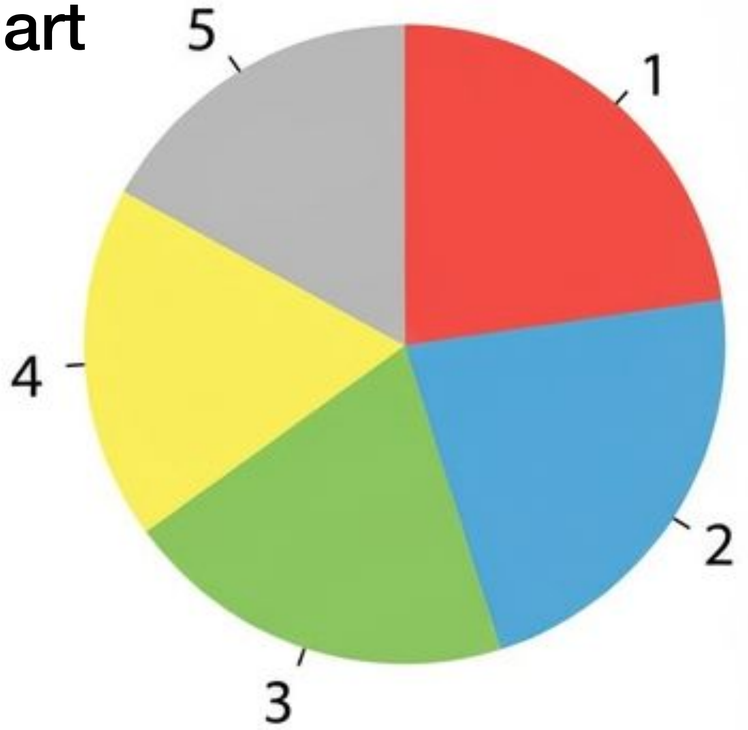
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course code:
WSTAPX

Which qualitative visual encoding is used?

Pie chart



(A) Position

(B) Color

(C) Shape

(D) Angle

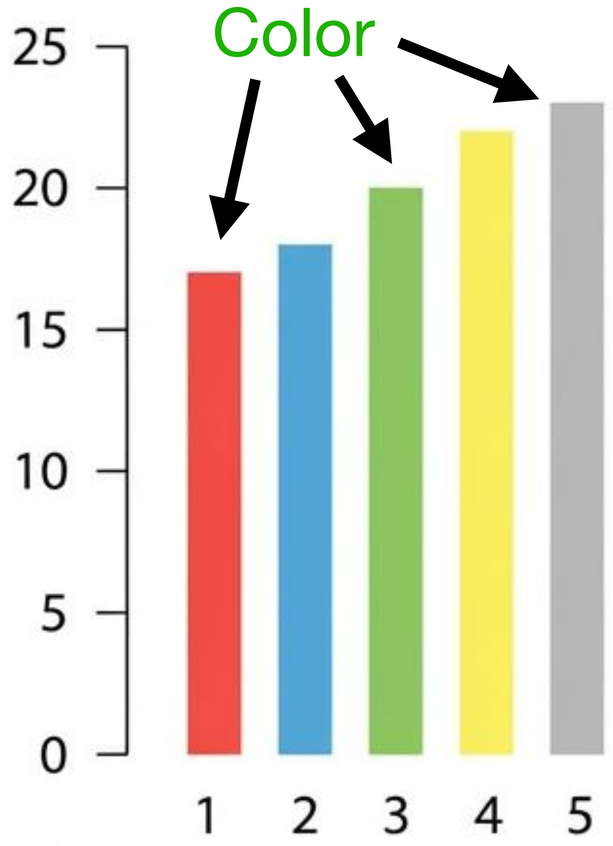
(E) Texture

Visual encodings

Quantitative encodings: *magnitude*

From **most to least** precise:

1. **Position**
2. **Length**
3. **Angle**
4. **Area**
5. **Color**



Bar plot

Length of bars encodes the values

Position on x-axis indicates different categories

Qualitative encodings: *categories*

1. **Position**
2. **Color**
3. **Shape**
4. **Texture/pattern**

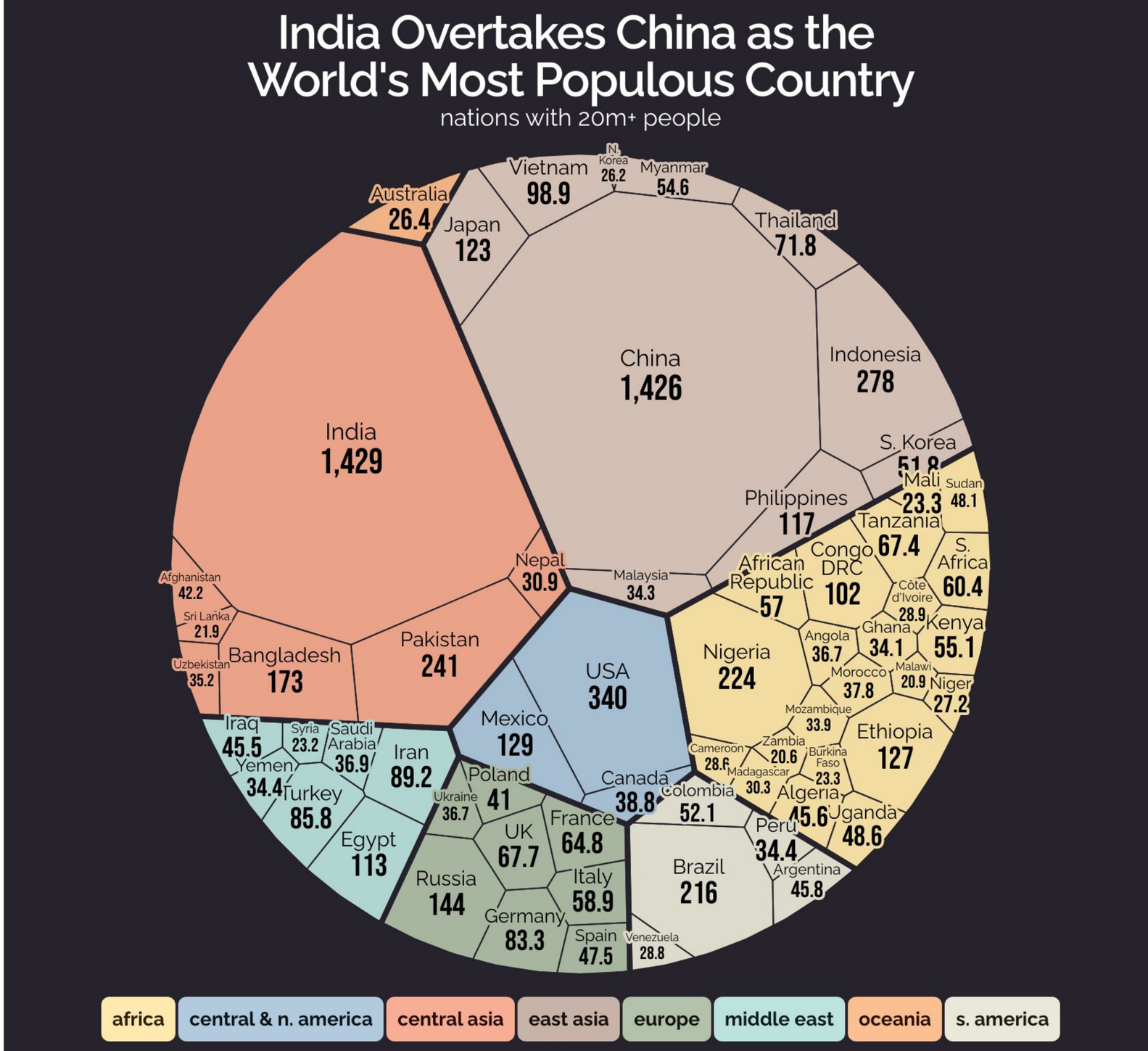
What makes a visualization effective?

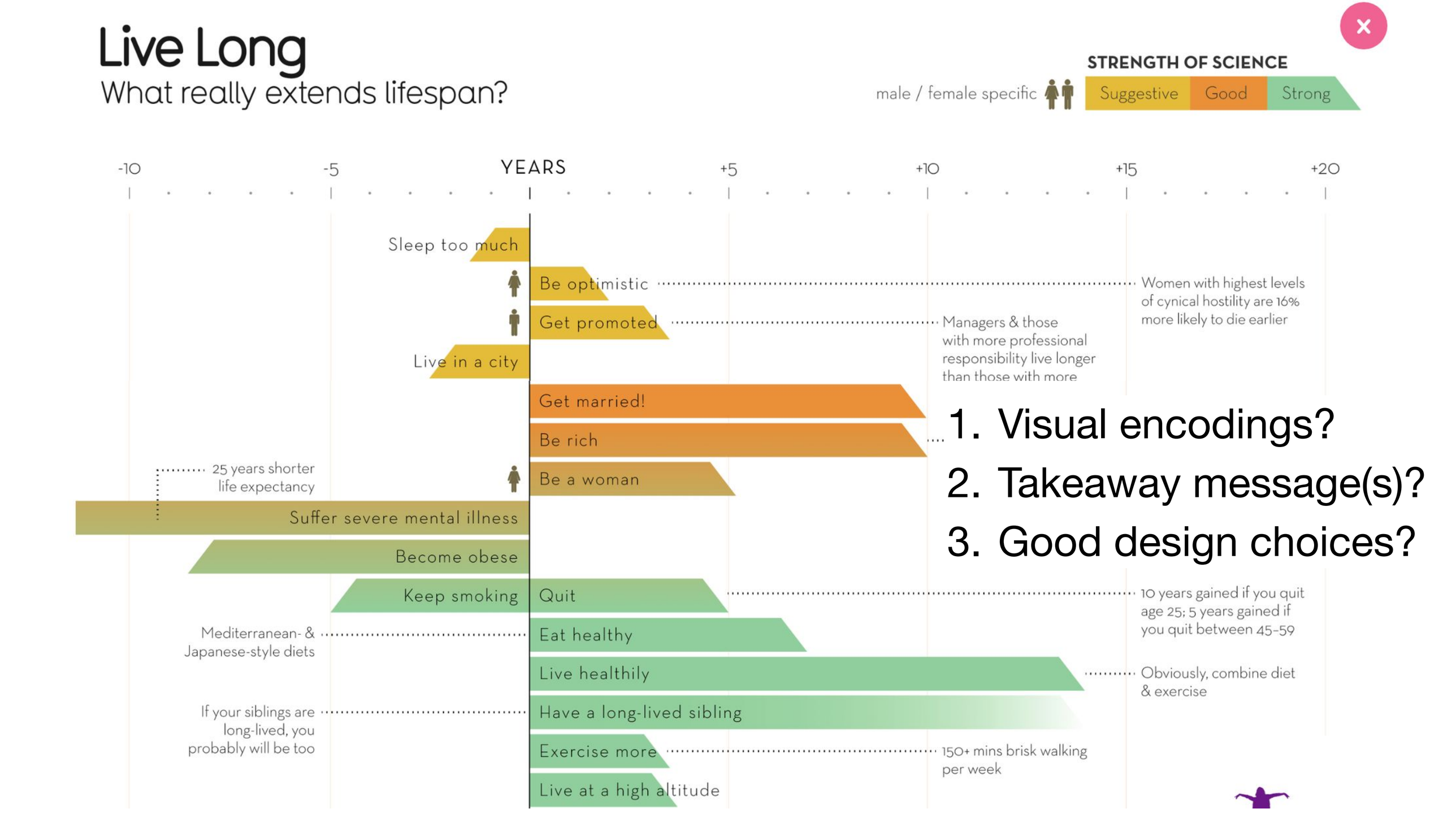
(~1 min with your neighbor)

For each of the following visualizations…

1. What visual encodings are used?
2. Takeaway message(s) from this plot?
3. What design choices help you understand the plot easily?

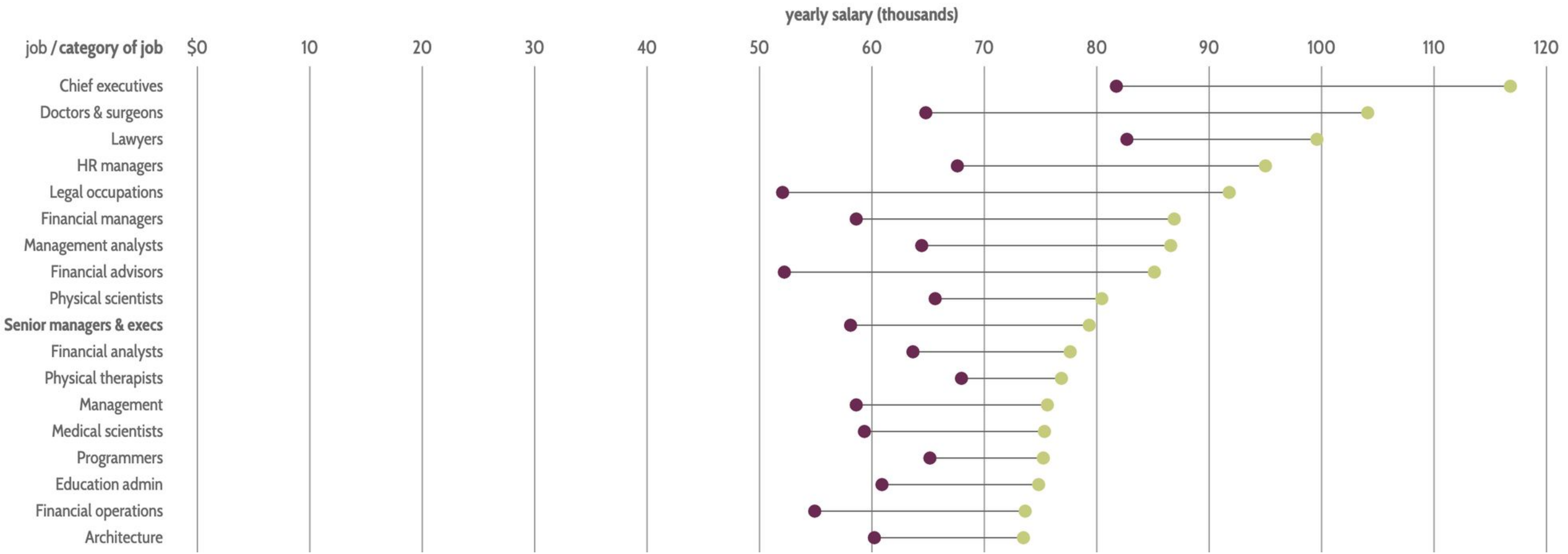
1. Visual encodings?
2. Takeaway message(s)?
3. Good design choices?





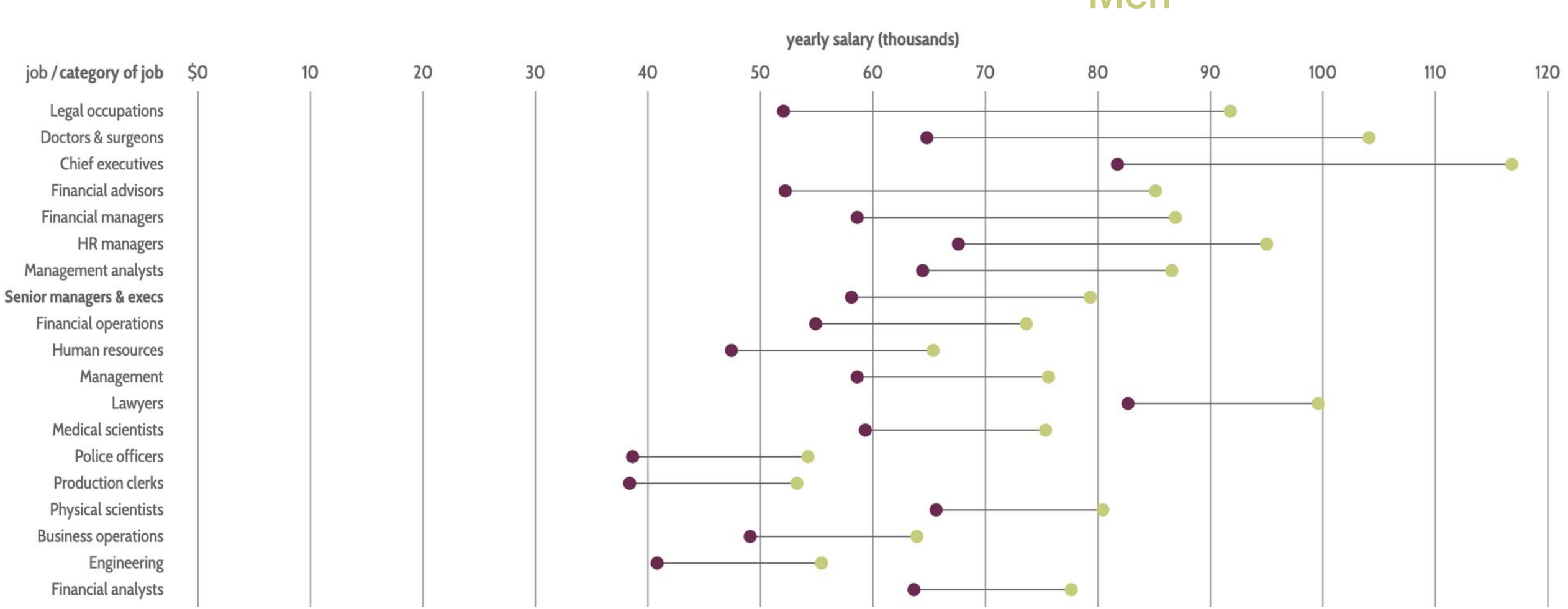
1. Visual encodings?
2. Takeaway message(s)?
3. Good design choices?

The Gender Pay Gap



1. Visual encodings?
2. Takeaway message(s)?
3. Good design choices?

The Gender Pay Gap



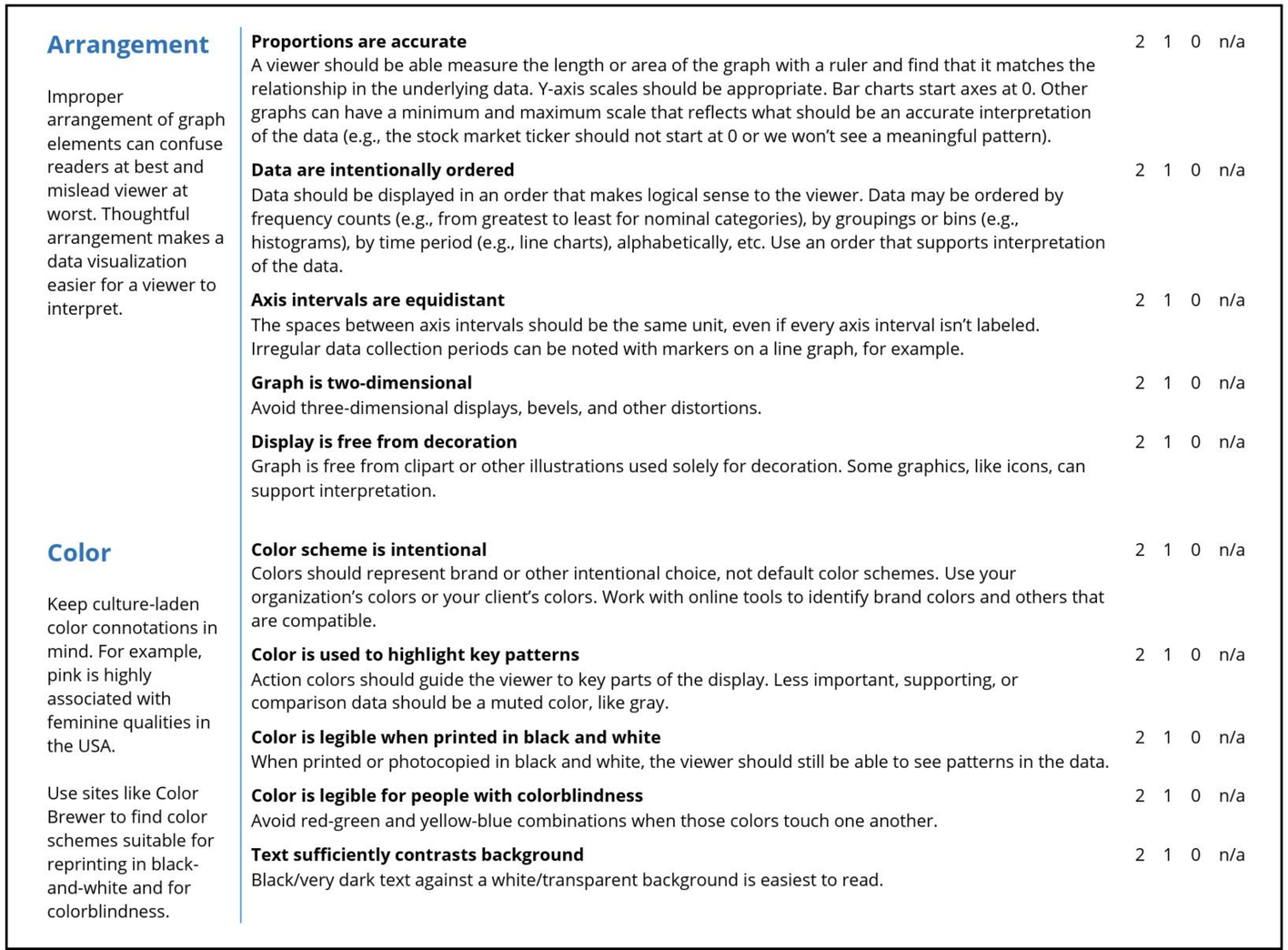
What does this version of the plot highlight that the other version did not?

Data Visualization Checklist

24 items regarding a visualization's

datavisualizationchecklist.com

1. Text/labels
2. Arrangement
3. Color choices
4. Lines and axes
5. Overall plot choice and message



Arrangement

Improper arrangement of graph elements can confuse readers at best and mislead viewer at worst. Thoughtful arrangement makes a data visualization easier for a viewer to interpret.

Proportions are accurate 2 1 0 n/a
A viewer should be able measure the length or area of the graph with a ruler and find that it matches the relationship in the underlying data. Y-axis scales should be appropriate. Bar charts start axes at 0. Other graphs can have a minimum and maximum scale that reflects what should be an accurate interpretation of the data (e.g., the stock market ticker should not start at 0 or we won't see a meaningful pattern).

Data are intentionally ordered 2 1 0 n/a
Data should be displayed in an order that makes logical sense to the viewer. Data may be ordered by frequency counts (e.g., from greatest to least for nominal categories), by groupings or bins (e.g., histograms), by time period (e.g., line charts), alphabetically, etc. Use an order that supports interpretation of the data.

Axis intervals are equidistant 2 1 0 n/a
The spaces between axis intervals should be the same unit, even if every axis interval isn't labeled. Irregular data collection periods can be noted with markers on a line graph, for example.

Graph is two-dimensional 2 1 0 n/a
Avoid three-dimensional displays, bevels, and other distortions.

Display is free from decoration 2 1 0 n/a
Graph is free from clipart or other illustrations used solely for decoration. Some graphics, like icons, can support interpretation.

Color

Keep culture-laden color connotations in mind. For example, pink is highly associated with feminine qualities in the USA.

Use sites like Color Brewer to find color schemes suitable for reprinting in black-and-white and for colorblindness.

Color scheme is intentional 2 1 0 n/a
Colors should represent brand or other intentional choice, not default color schemes. Use your organization's colors or your client's colors. Work with online tools to identify brand colors and others that are compatible.

Color is used to highlight key patterns 2 1 0 n/a
Action colors should guide the viewer to key parts of the display. Less important, supporting, or comparison data should be a muted color, like gray.

Color is legible when printed in black and white 2 1 0 n/a
When printed or photocopied in black and white, the viewer should still be able to see patterns in the data.

Color is legible for people with colorblindness 2 1 0 n/a
Avoid red-green and yellow-blue combinations when those colors touch one another.

Text sufficiently contrasts background 2 1 0 n/a
Black/very dark text against a white/transparent background is easiest to read.

How do I choose the right visualization for my data?